

KICKOFF SESSION · FALL 2026

Senior Capstone Project

CS 480W / CS 481 — From Concept to Deployment in Two Semesters

Professor Fred Agbo, PhD

Assistant Professor of Computer Science

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01

Welcome to Your Capstone

This is the culmination of everything you've learned — where you design, build, and ship something real.



1

Two full semesters

Fall 2026 (CS 480W) through Spring 2027 (CS 481) — 8 credits, one continuous project.

2

Your own project, start to finish

You choose it, scope it, architect it, build it, test it, and deploy it.

3

Industry-standard practice

Real SDLC methodology: agile sprints, code review, CI/CD, and documentation — the way professional teams work.

4

A portfolio piece that speaks for you

Something you can demo in an interview, not just describe on a resume.

Meet Your Instructor



Professor Fred Agbo, PhD

Assistant Professor of Computer Science



Ford Hall 209 · Salem



fjagbo@willamette.edu



Office hours: 15-min appointments or drop-in — in person, phone, or Google Meet

Think of me as your Product Owner.

I'll set direction, review your sprints weekly, and push your project toward something you're proud to ship — but the roadmap, the decisions, and the code are yours.

03

By the End, You Will Have...



A complete application

Full-stack, concept through deployment



Modern dev practices

Proven proficiency across the SDLC



Real documentation

Technical docs that outlast the course



Presentation skill

For technical and general audiences alike



A portfolio piece

Proof of what you can actually build

The Journey Ahead — Two Semesters, One Project

SEMESTER 1 • CS 480W • FALL 2026

Weeks 1–4

Initiation &
Planning

Weeks 5–8

Design &
Architecture

Weeks 9–14

Foundation
Development

Week 15

Wrap-Up &
Prototype Demo

SEMESTER 2 • CS 481 • SPRING 2027

Weeks 1–6

Feature
Development

Weeks 7–10

Testing &
Quality Assurance

Weeks 11–13

Deployment &
Documentation

Weeks 14–15

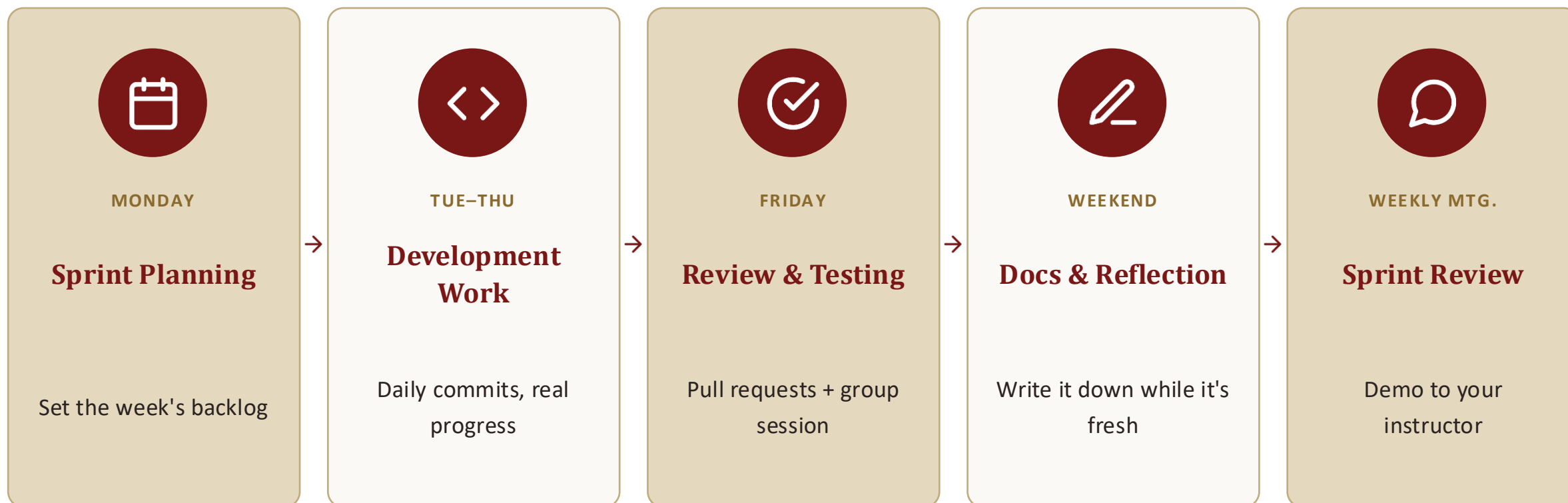
Showcase &
Handoff

Full roadmap and weekly deliverables are in the syllabus — this is your flight path for the year.

05

Your Weekly Rhythm

Every week runs on the same cycle — consistency is what makes agile work.



Two Meetings, Every Single Week



Group Review Session

Every Friday · 10:30–11:00 AM · Ford Hall 209

- Attendance expected from everyone
- Review the week's milestones together
- Peer collaboration & shared problem-solving



Individual Sprint Review

15 minutes, weekly · You book it

- You reserve your own slot each week
- Demo your progress 1-on-1
- Personalized feedback on blockers



Book Your Sprint Review — link on course site

How We Run Sprints: Agile Ceremonies



Sprint Planning

Weekly · 10 min

Review velocity, select
backlog items, commit to
goals



Daily Standup

Async, by 10 AM

Post in Discord/Slack:
yesterday, today, blockers



Sprint Review

Weekly · 15 min

Demo features, gather
feedback, discuss what's left



Retrospective

Bi-weekly · 5 min

What went well, what to
improve, action items

Non-Negotiables: Development Practices



Version Control

- Daily commits — minimum, including doc days
- Conventional commits: feat:, fix:, docs:
- Feature-branch workflow with pull requests
- Main branch is protected — requires review



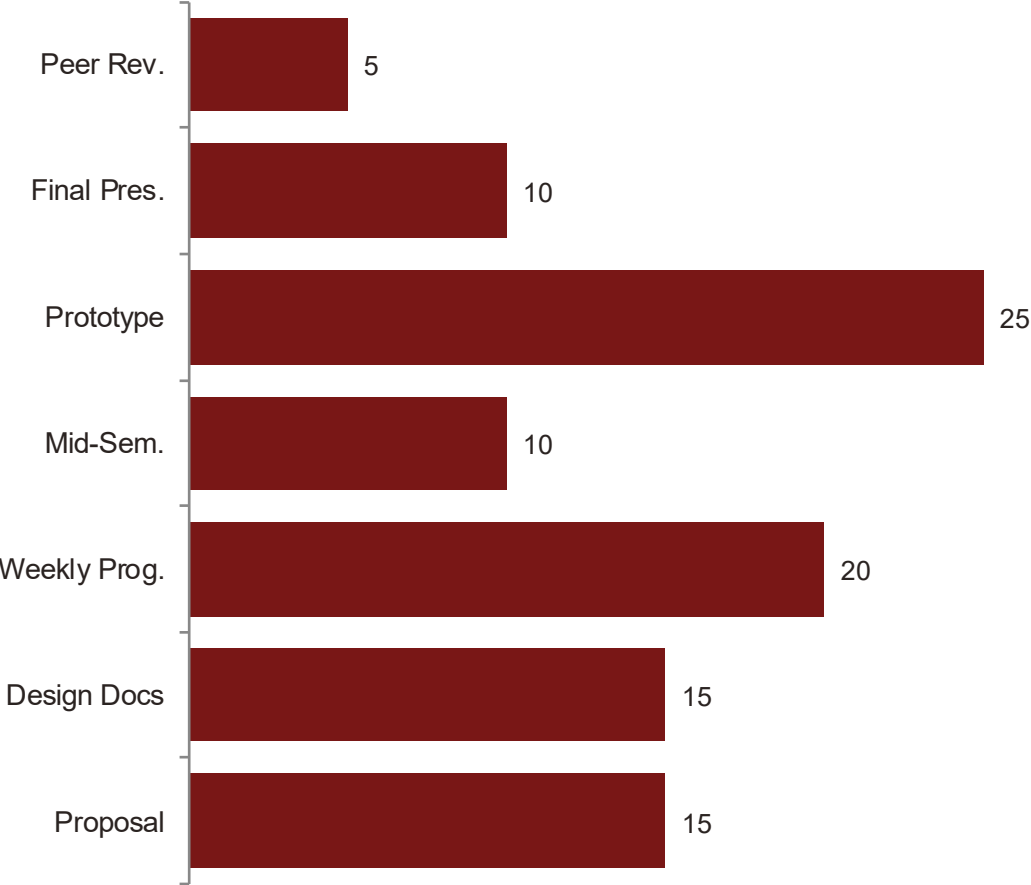
Testing & CI/CD

Unit tests	60% min (Sem. 2)
Integration tests	100% of API endpoints
End-to-end tests	Core user paths
Performance	< 100ms query response

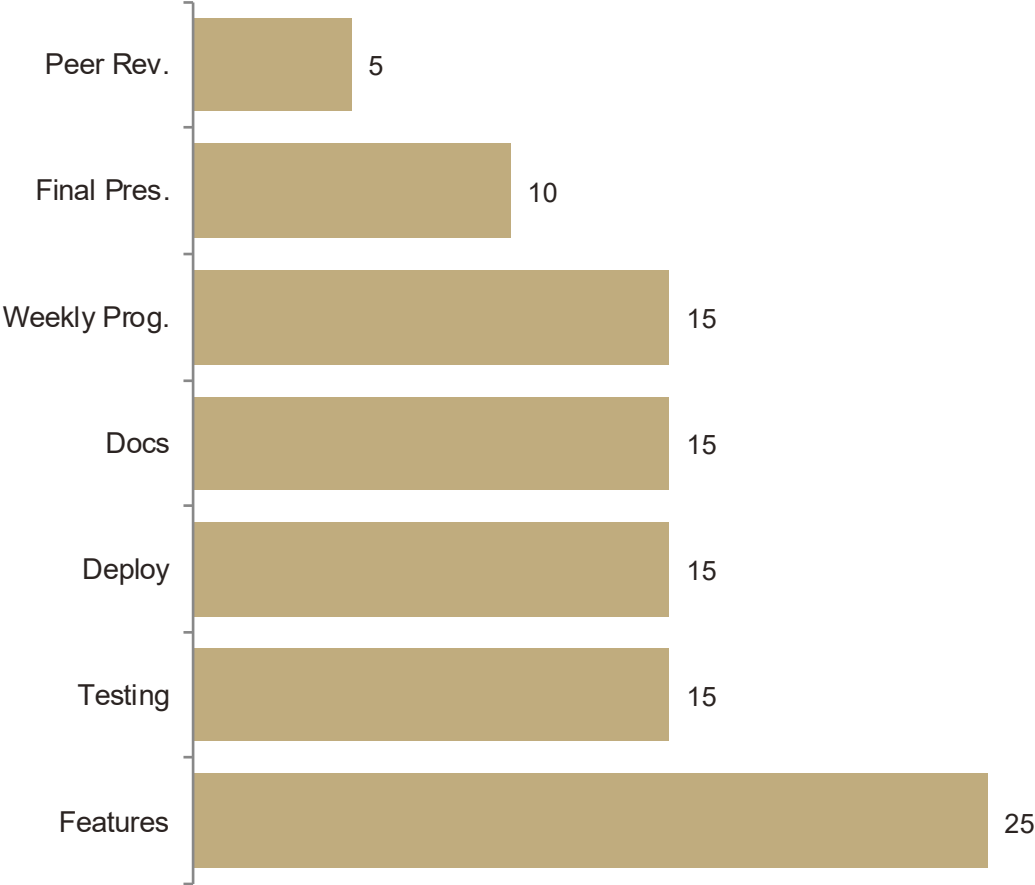
CI/CD pipeline required — GitHub Actions, Jenkins, or CircleCI

How You're Graded, Semester by Semester

Semester 1 · CS 480W



Semester 2 · CS 481



You're Also a Reviewer

You'll review one classmates' projects across both semesters — same partners throughout, real critical-evaluation practice.



**Assigned once,
consistent all year**

Review Rubric

A • 90–100%

Specific, actionable feedback with code examples

B • 80–89%

Identifies major issues with useful suggestions

C • 70–79%

Meets requirements with basic feedback

D • 60–69%

Minimal effort, vague feedback

F • Below 60%

Missing or non-constructive

Policies Worth Knowing Cold



Time Commitment

6–9 hrs/week outside class — dev, docs, and project work combined.



Academic Integrity

All code original or attributed. AI tool usage must be documented.



Late Work

72 cumulative hours free, then 20% penalty per 24 hrs. No credit past 5 days.



Incomplete Grades

Reserved for prolonged illness or family emergency — not falling behind.

What Actually Predicts Success

1

Start early and iterate often

2

Commit code daily, even small changes

3

Ask for help when stuck more than 2 hours

4

Attend every weekly meeting prepared

5

Engage meaningfully in peer reviews

6

Document as you go, not at the end

Before Next Friday...



Read the full syllabus

Course policies, grading, and the complete milestone schedule



Book your first Sprint Review

Reserve a weekly 15-minute slot on Google Calendar



Set up your dev environment

Git 2.0+, and Docker 20.0+ recommended



Draft three project ideas

One page each — due Week 1, the spark for everything ahead